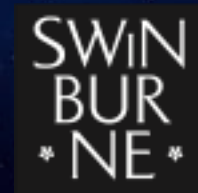


Discovering the Origins of Gravitational-Wave Sources (Heavy Double Neutron Star edition)

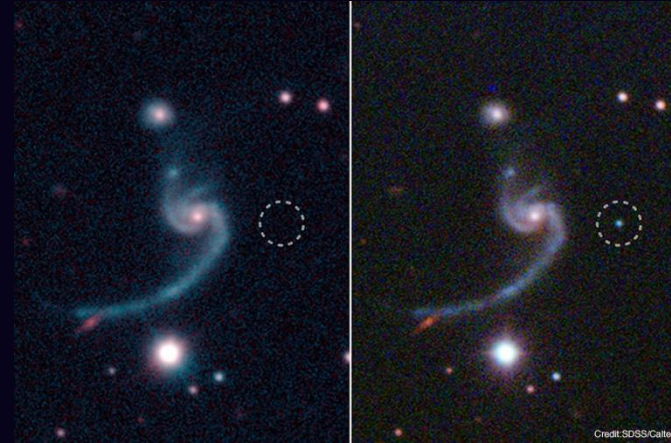
Primary Supervisor : Dr. Simon Stevenson
Secondary Supervisor : Prof. Jarrod Hurley



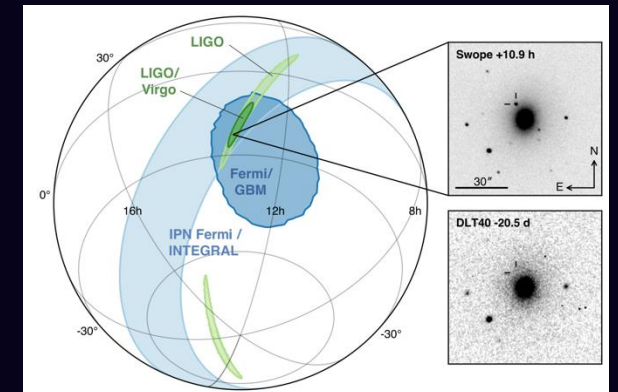
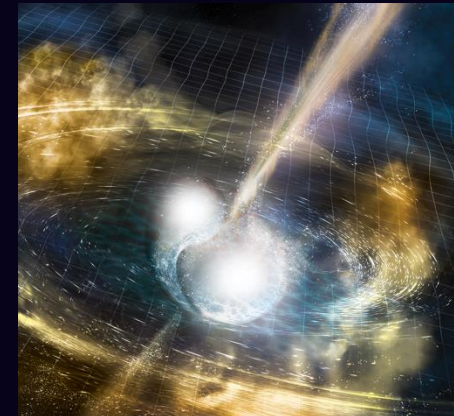
Centre for
Astrophysics and
Supercomputing

Why care about double neutron stars?

- Supernovae explosion (Sne)
- gamma ray burst (GRB) and kilonovae
- Formation of heavy r-process elements
- Late binary stellar evolution



Left: Red and green composite image from the Sloan Digital Sky Survey (SDSS) taken before SN iPTF14gqr. Right: Red/green/blue composite image from Palomar 16-inch telescope taken on Oct 19, 2014. During SN iPTF14gqr. (Credit: SDSS/Caltech)



Gamma ray burst and Kilonova AT2017gfo in the galaxy NGC 4993 2secs after GW170817

What do we know about DNS systems so far ?

Detected by radio telescopes and gravitational-wave detectors

Radio Astronomy

~ 34 Galactic DNS between total mass of $2.50\text{--}2.89M_{\odot}$

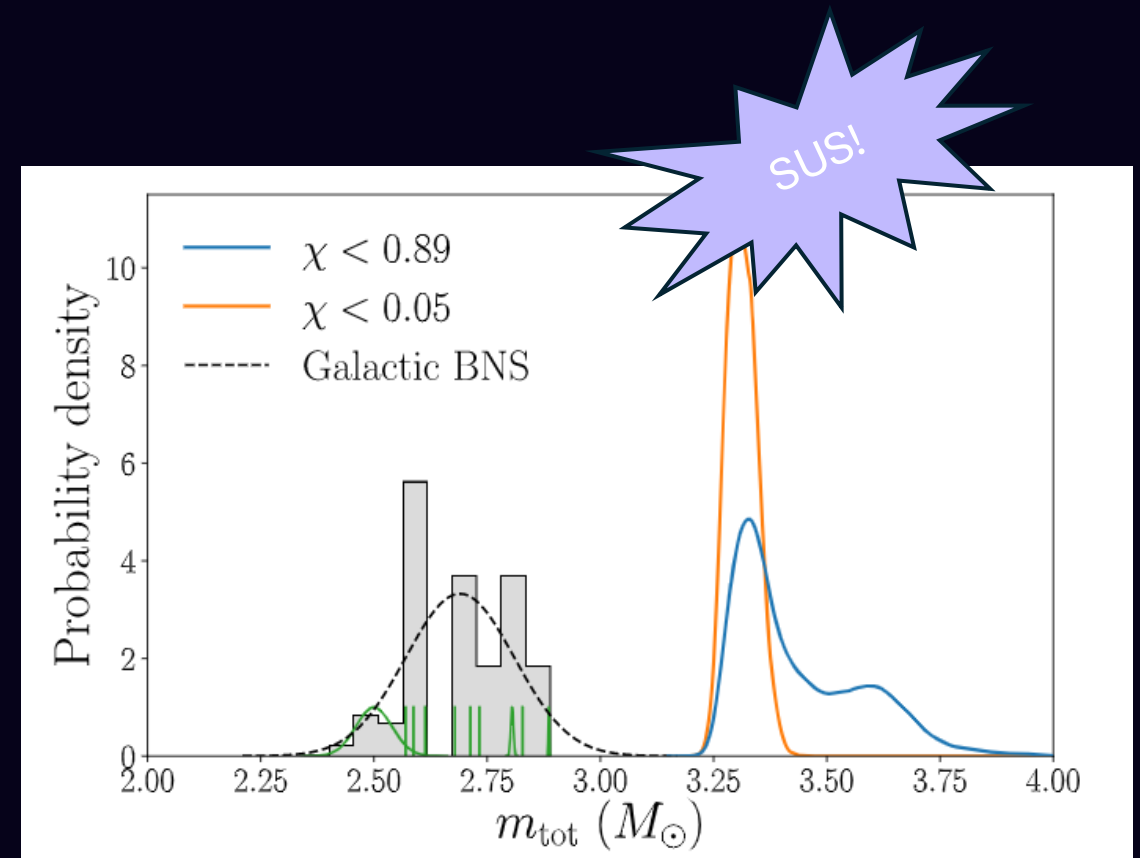
Gravitational Waves

GW170817

component masses : 1.17 to $1.60 M_{\odot}$
Total mass : $2.87^{+0.47}_{-0.09} M_{\odot}$

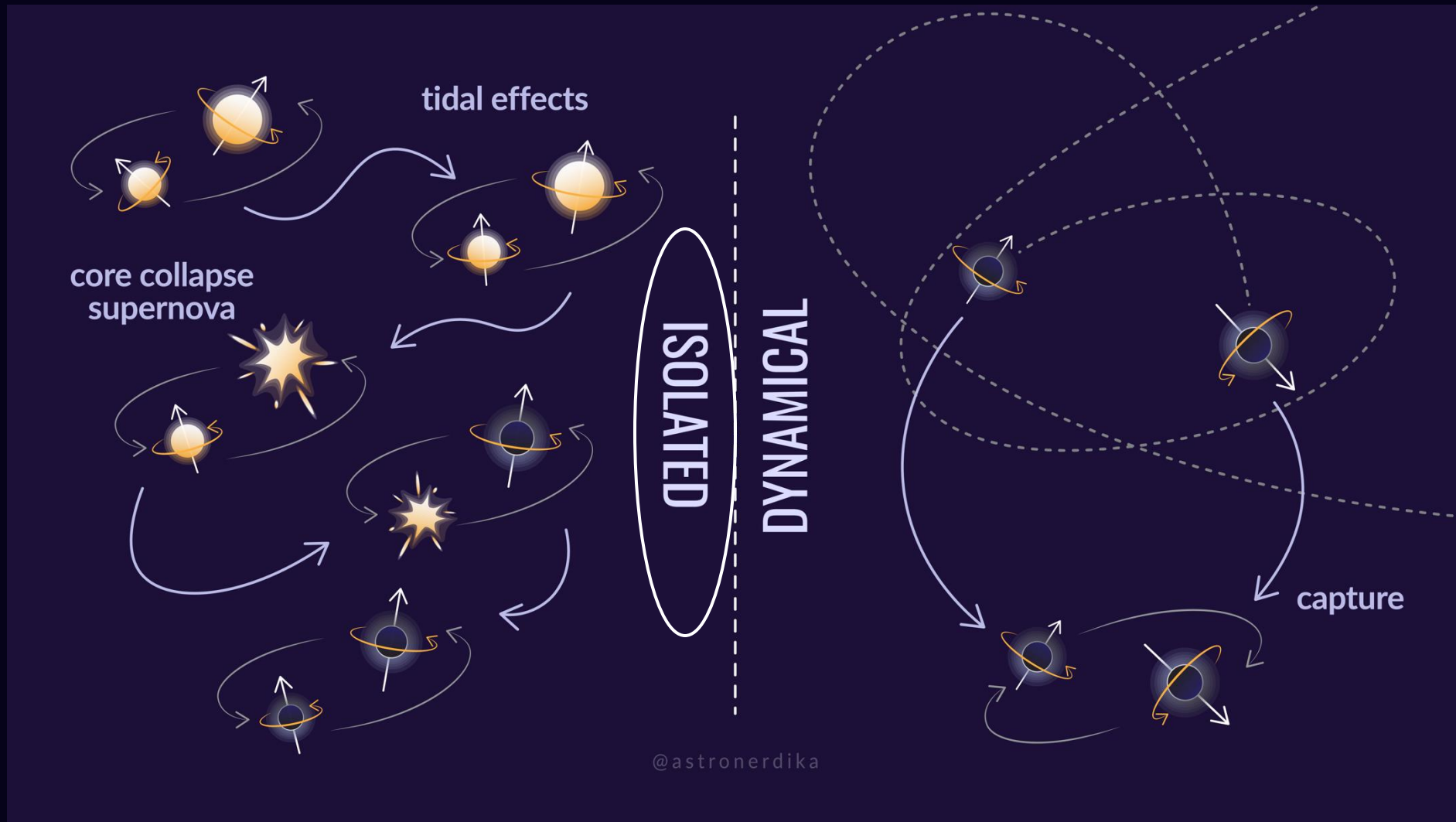
GW190425

component masses : 1.1 to $2.5 M_{\odot}$
Total mass : $3.4^{+0.3}_{-0.1} M_{\odot}$

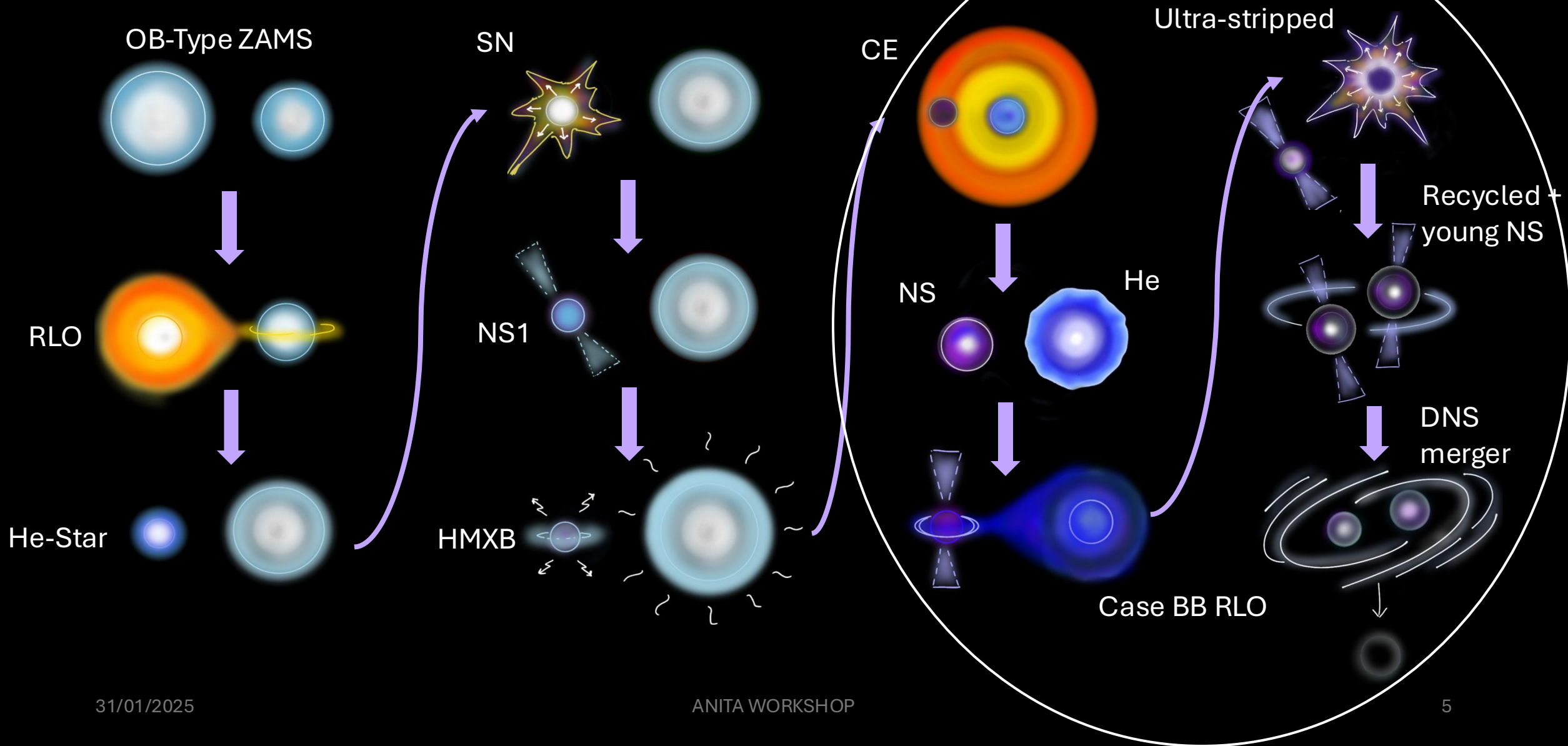


Abbott et al. 2020

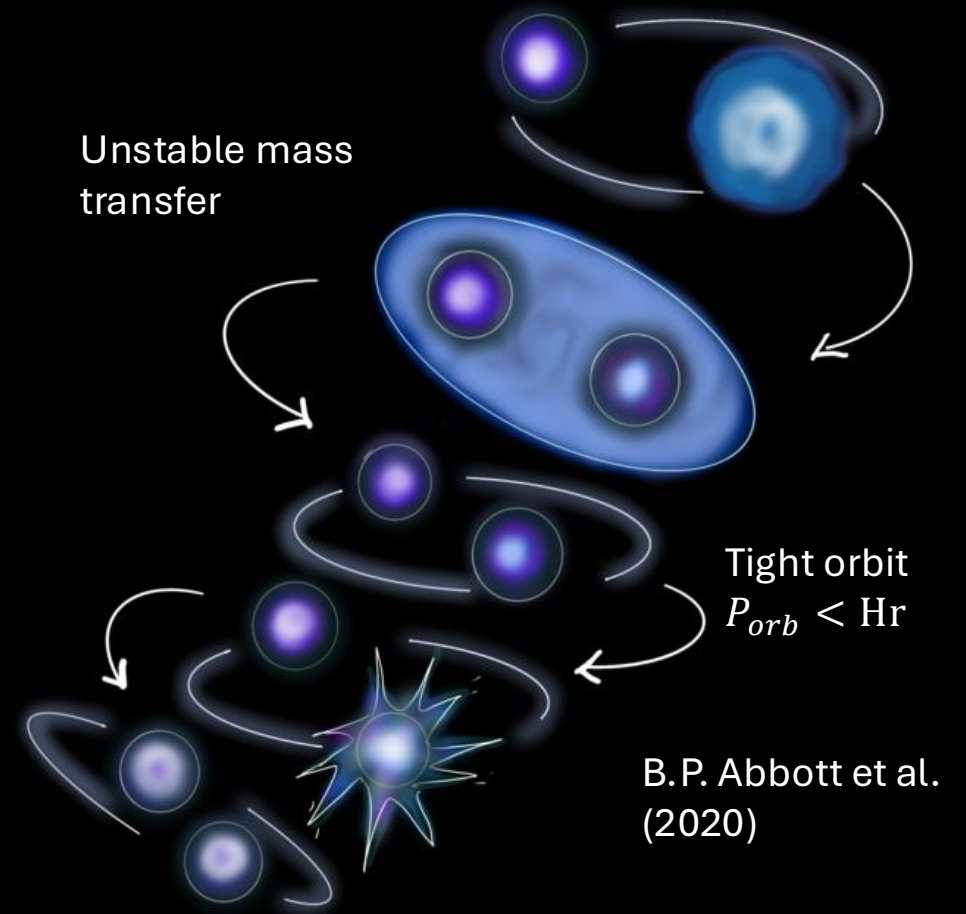
Formation Channels of compact binary mergers



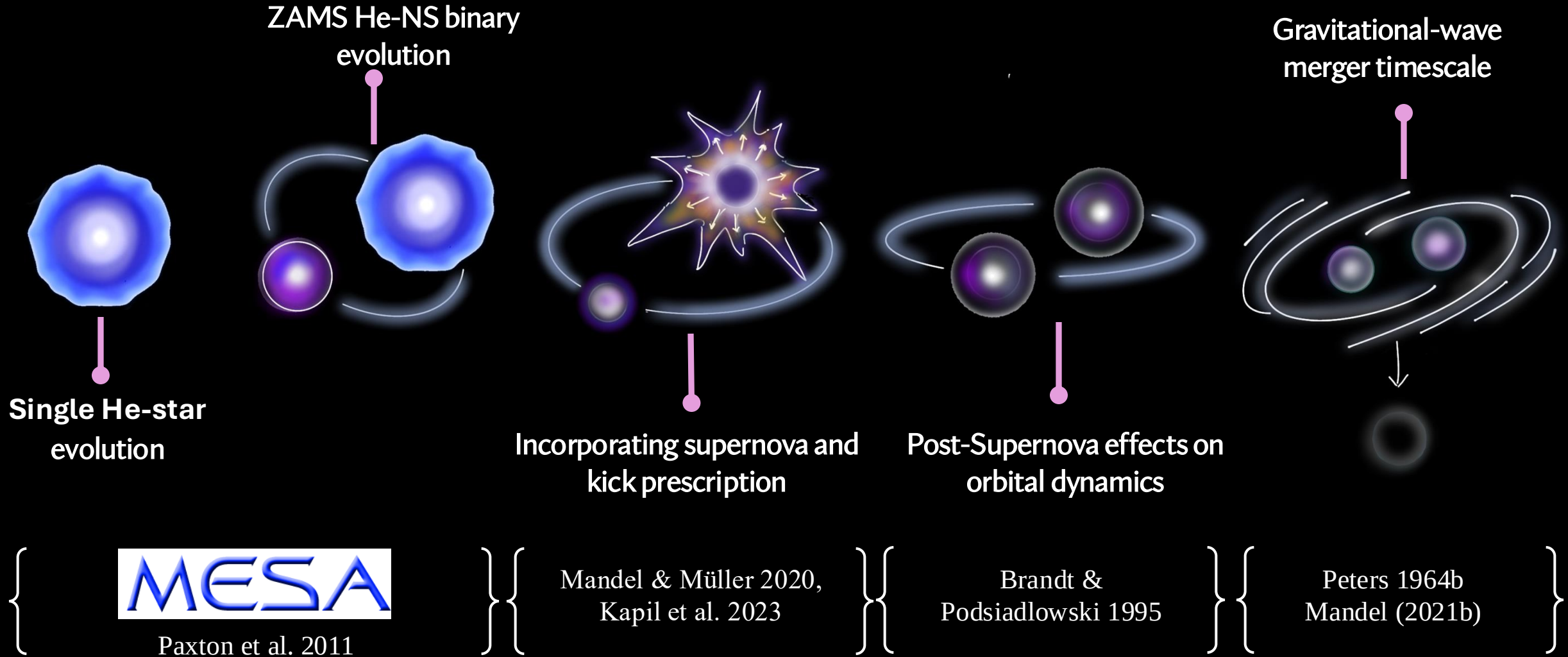
Standard Double Neutron Star formation pathway – T. Tauris et al. 2017



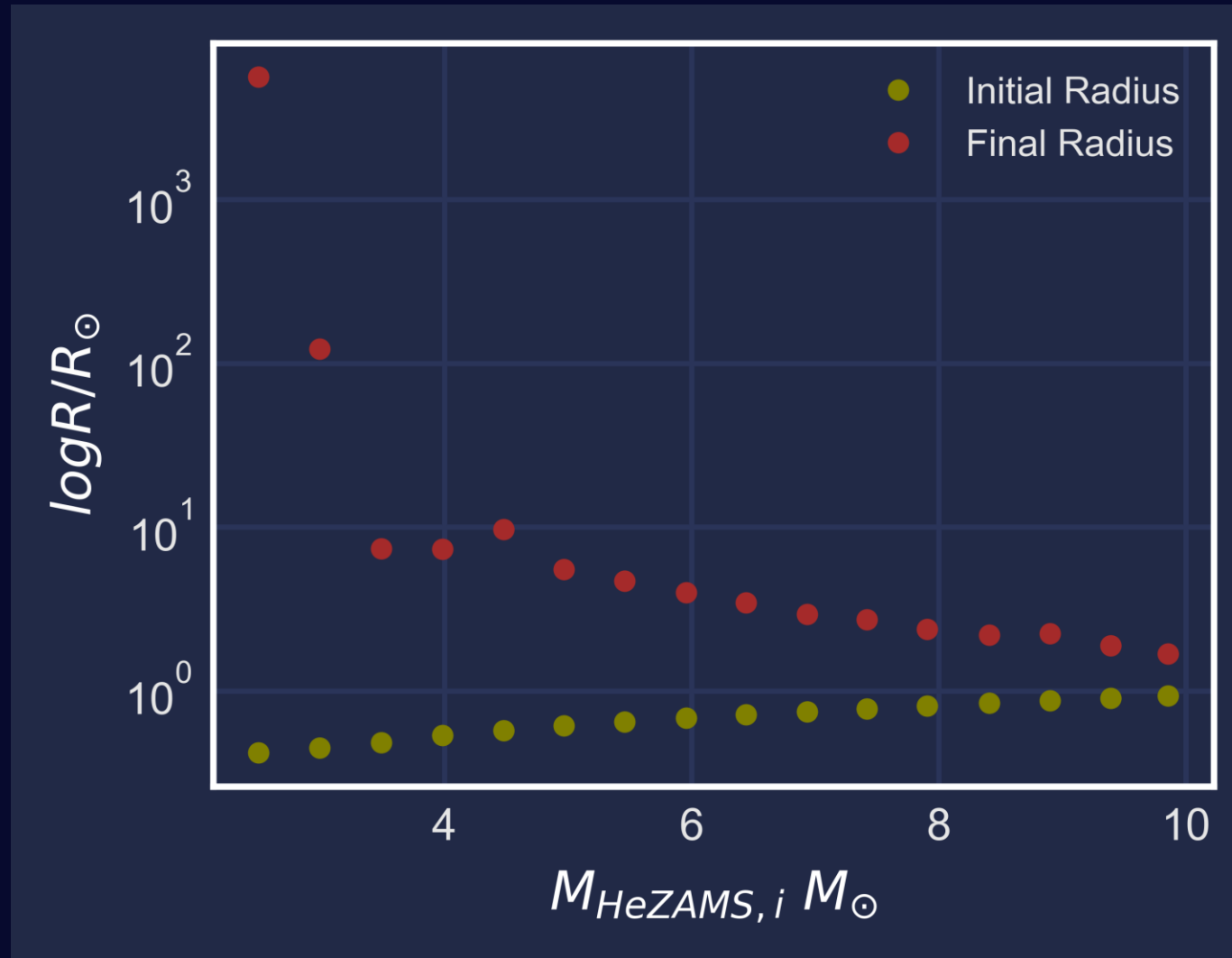
Alternate formation pathways



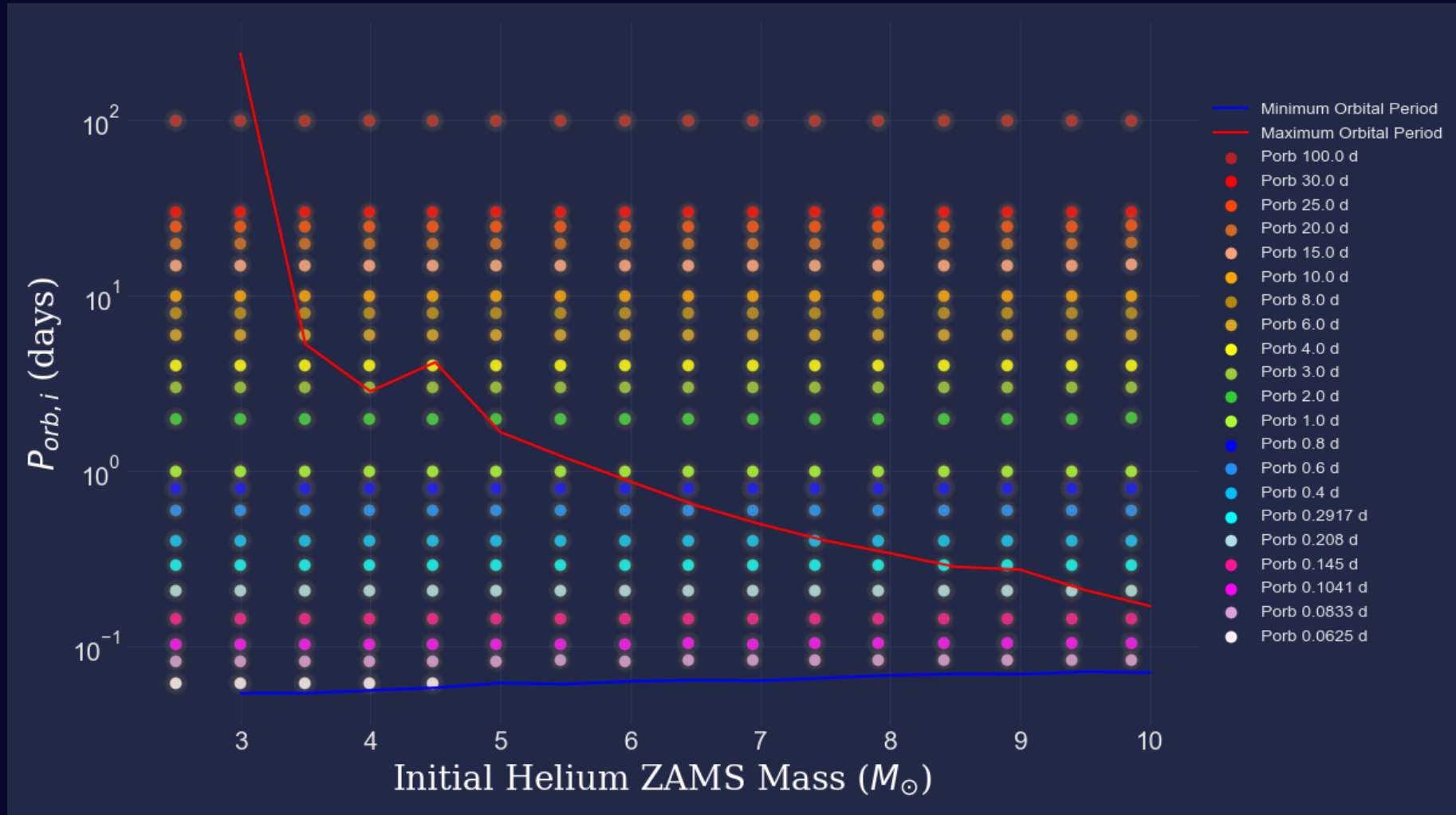
Framework created for the formation pathway of double neutron star



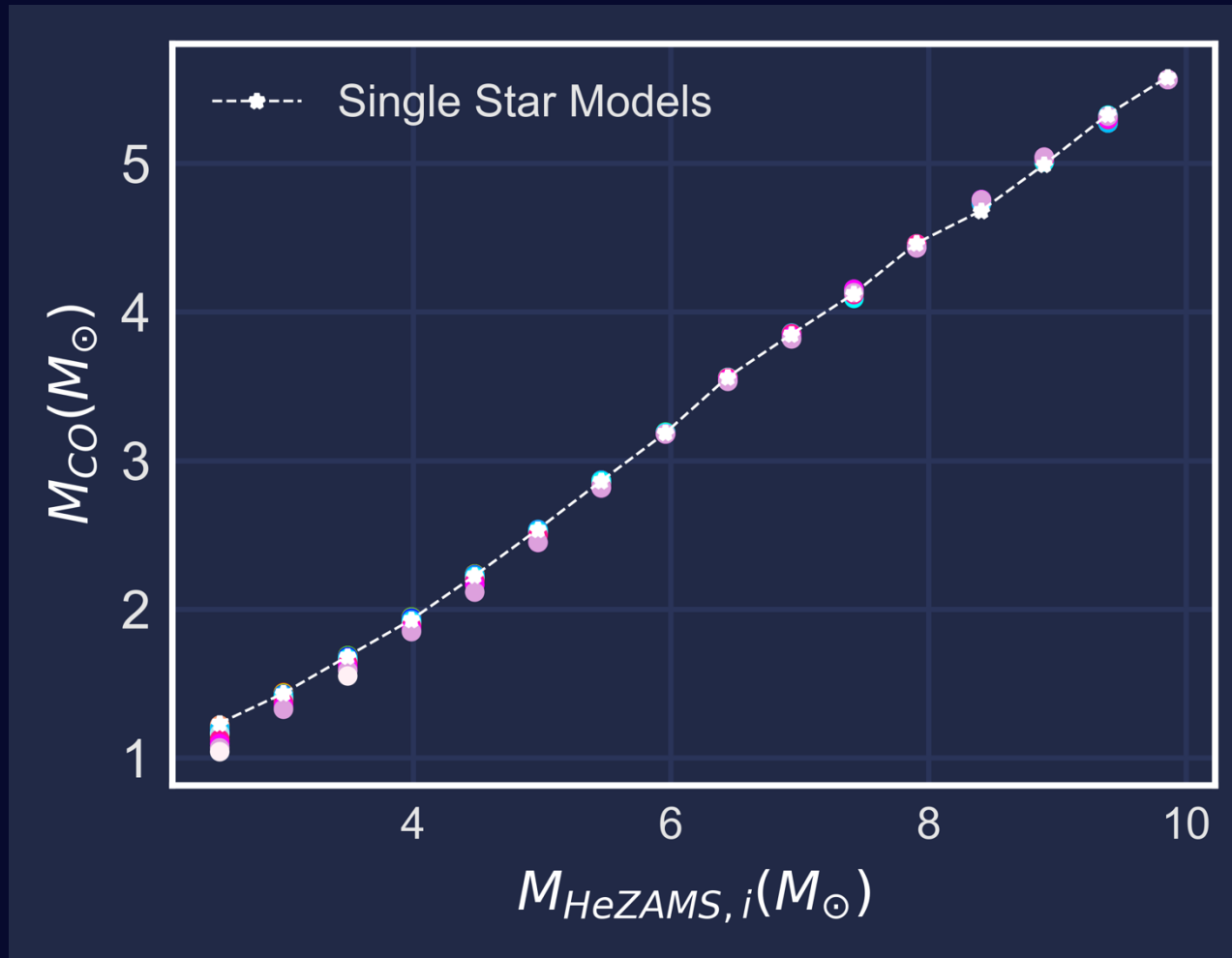
Initial and final radius as a function of initial He-ZAMS mass at solar metallicity



Grid of models based on initial He ZAMS mass + $1.4 M_{\odot}$ NS and initial orbital period

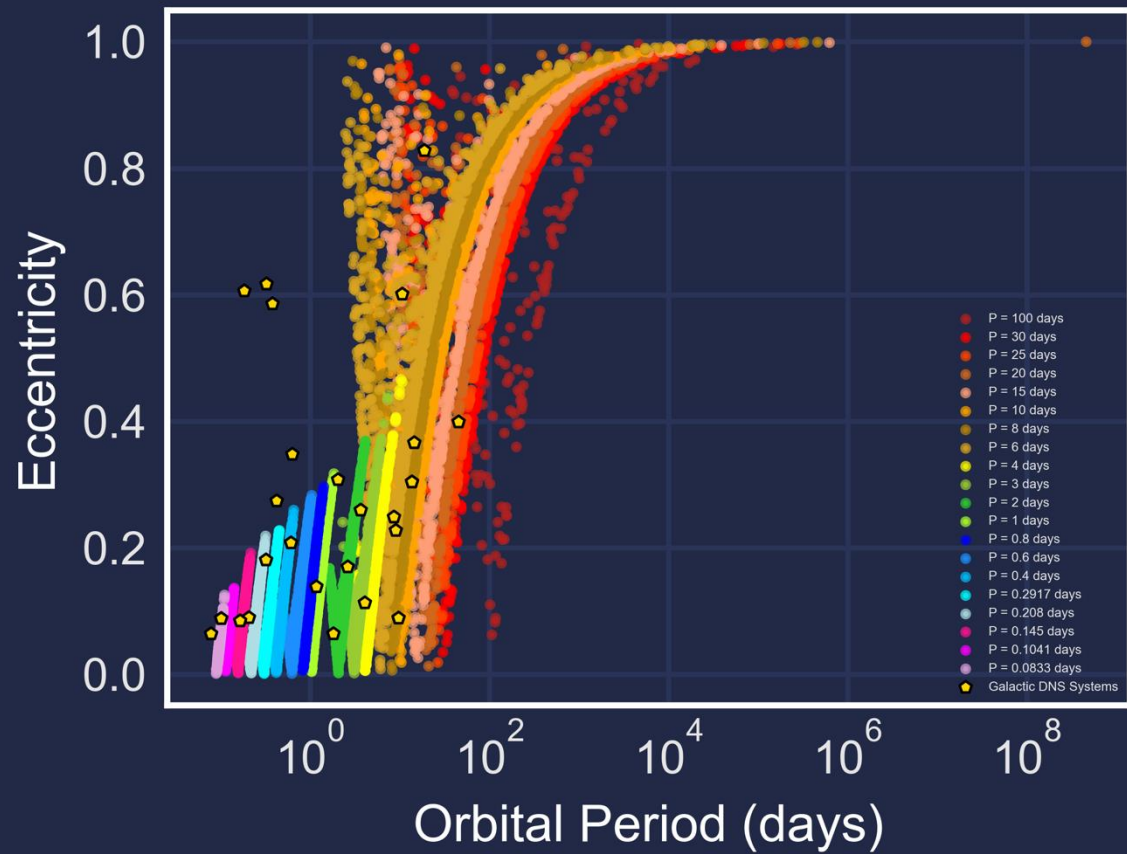


Carbon-oxygen (CO) core masses as function of initial He-ZAMS mass



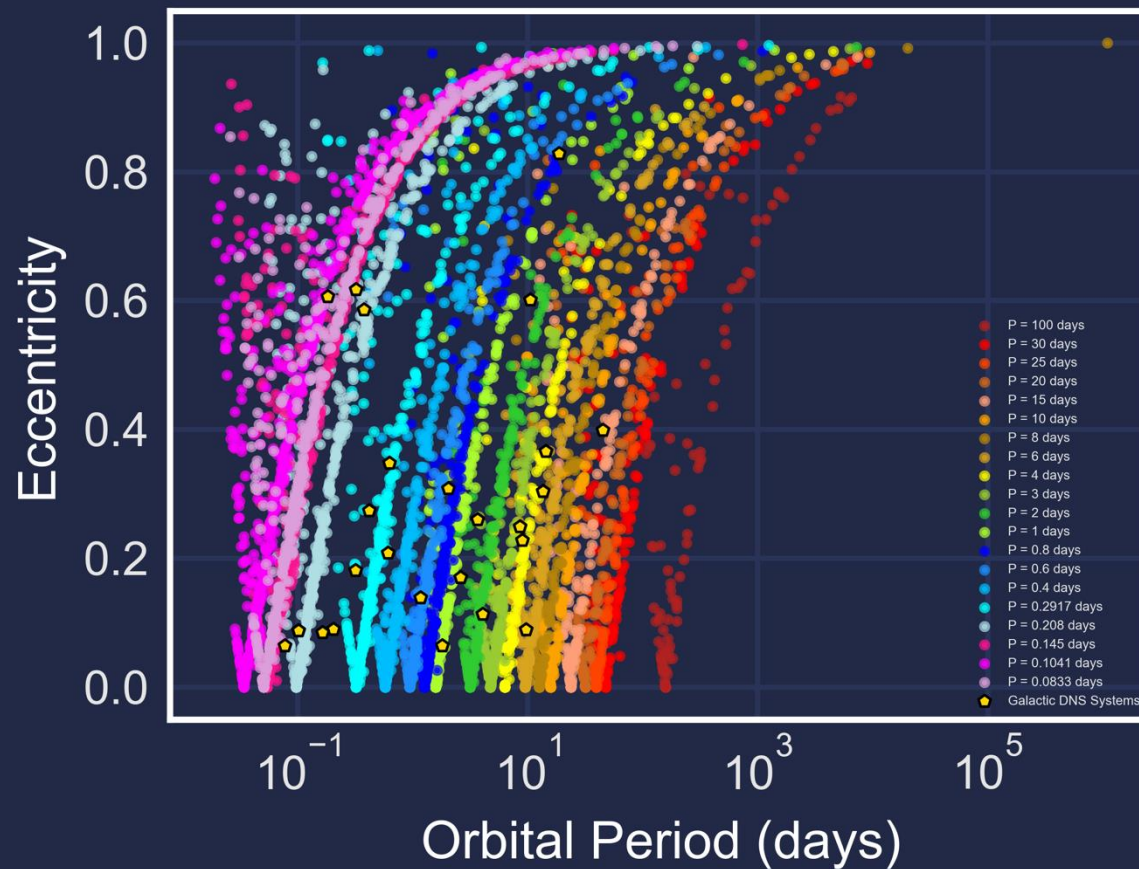
$$M_{He,i} = 2.9 M_{\odot} + 1.4 M_{\odot} \text{ NS},$$

$$M_{CO} = 1.33-1.44 M_{\odot}$$

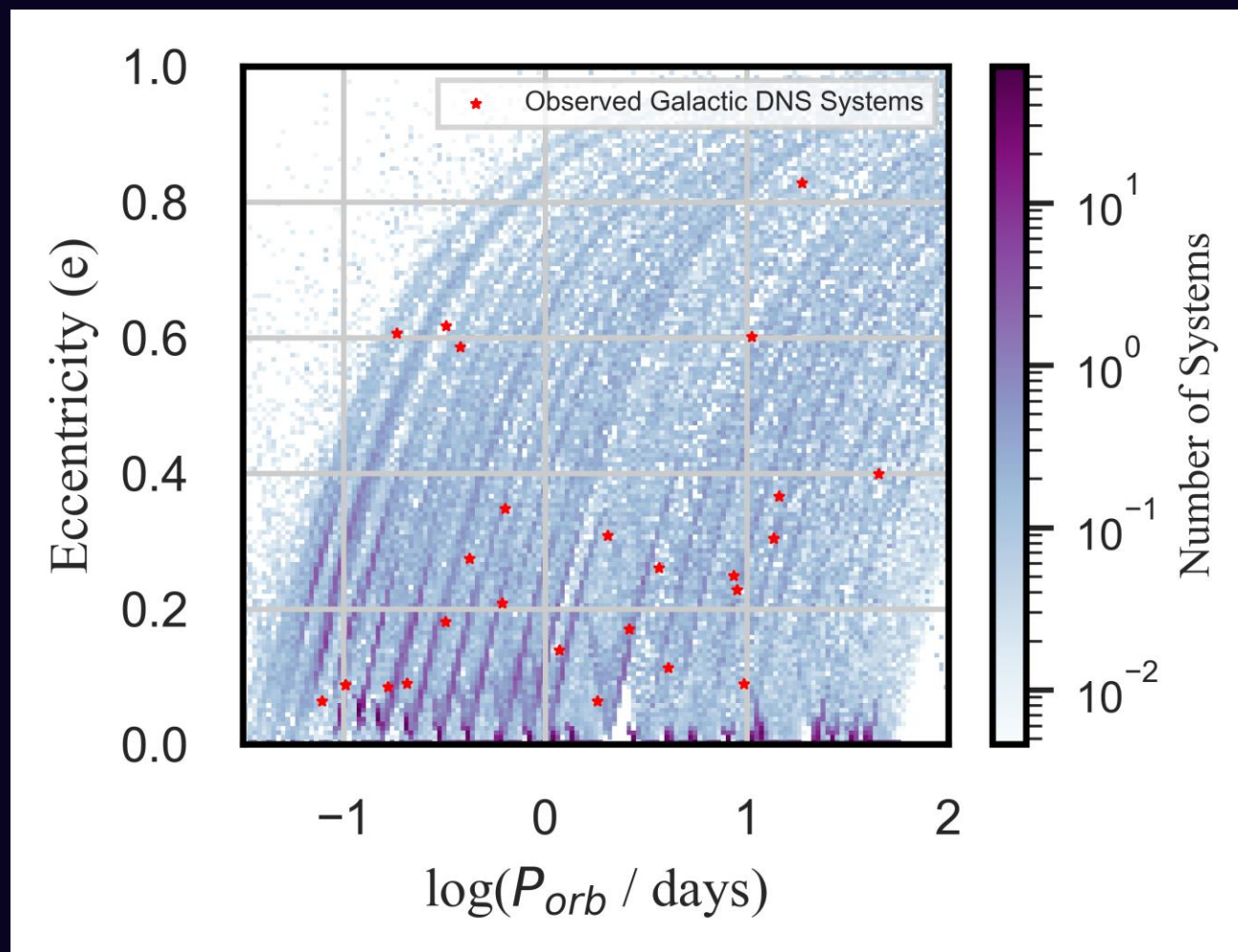


$$M_{He,i} = 5.9 M_{\odot} + 1.4 M_{\odot} \text{ NS},$$

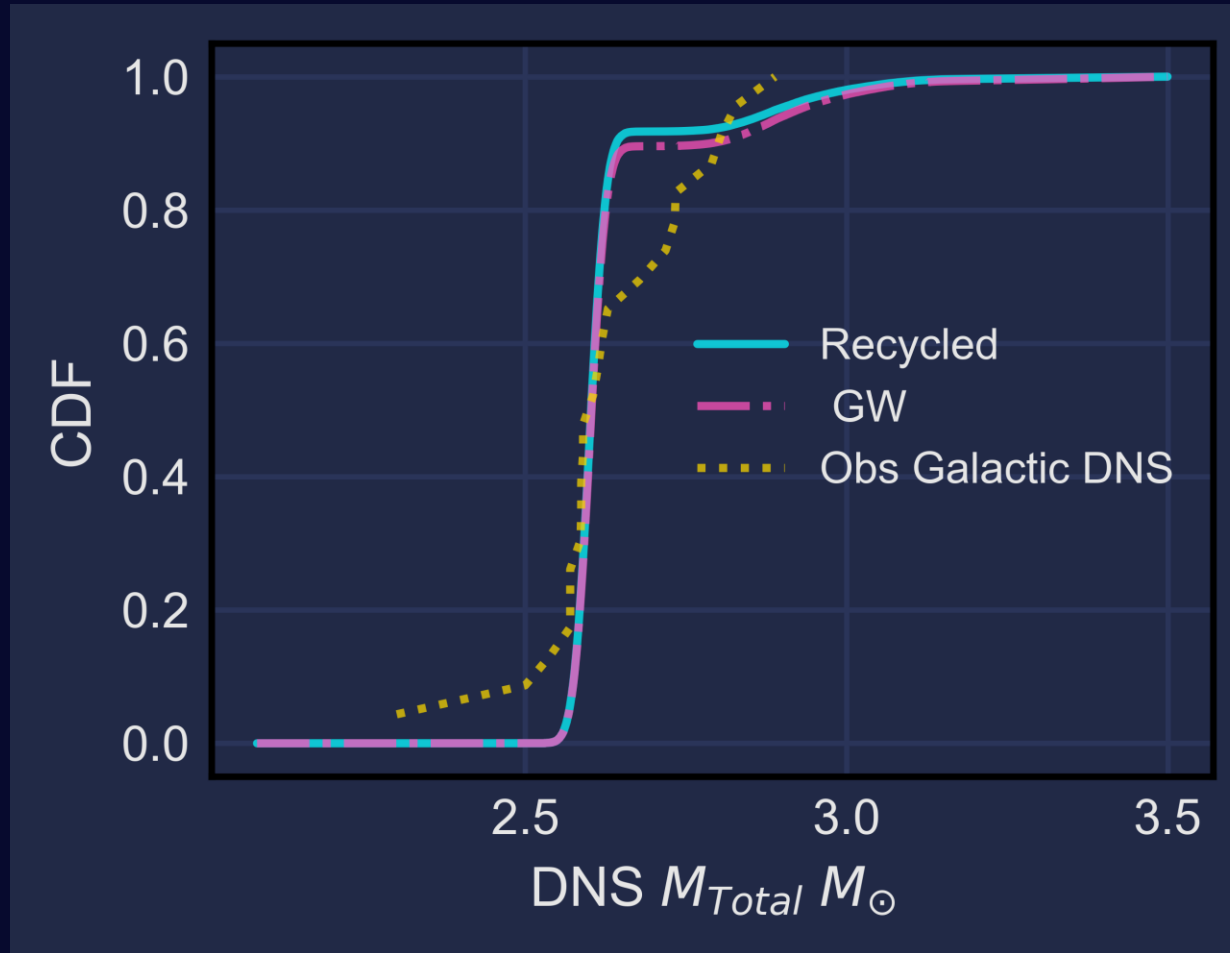
$$M_{CO} = 1.33-1.44 M_{\odot}$$



2D histogram of P_{orb} - e of all models



DNS Total mass distribution



Fraction of heavy DNS
($>3.0 M_{\odot}$)

GW = 8.83%

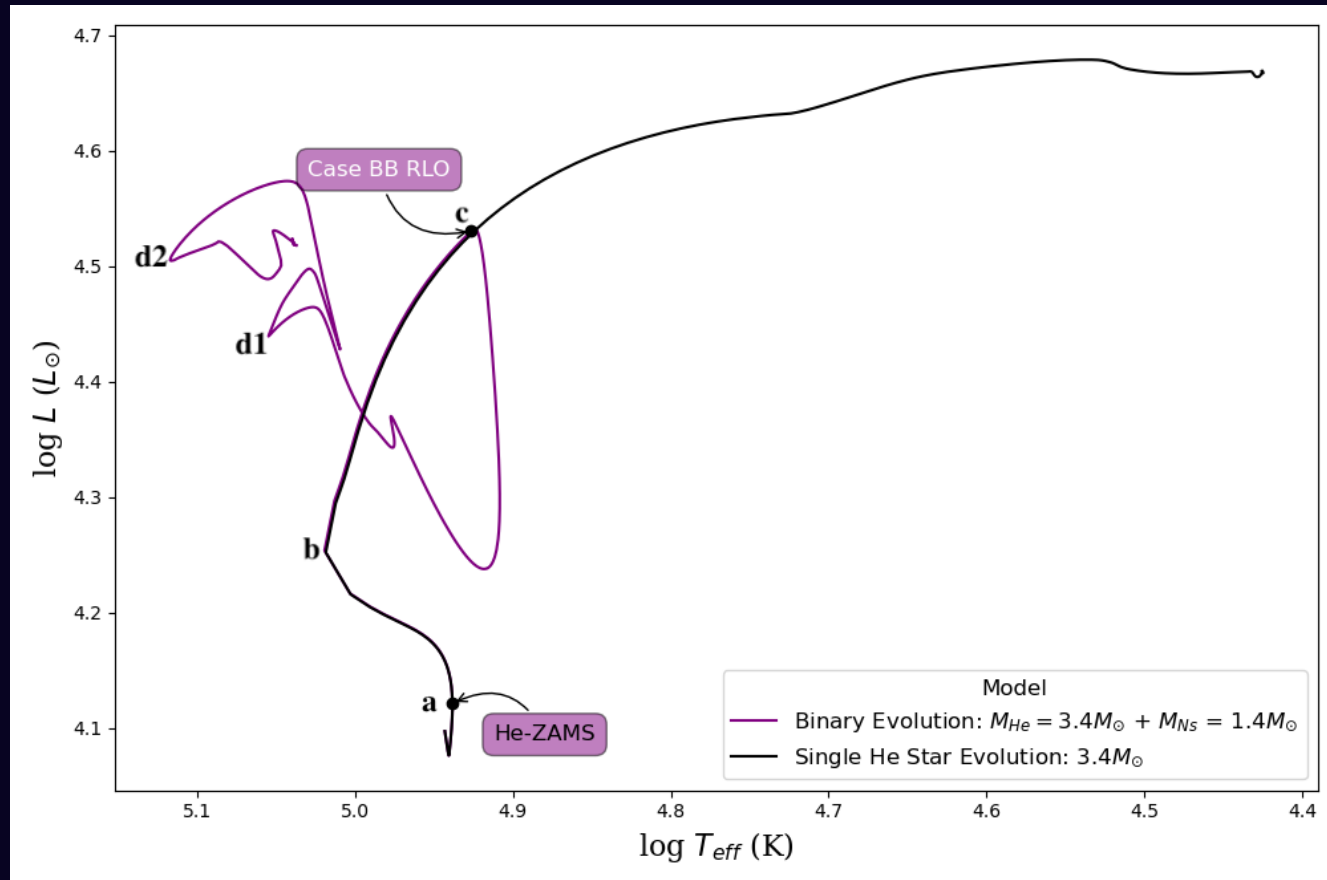
Recycled (radio) = 7.24%

Conclusion and key take aways

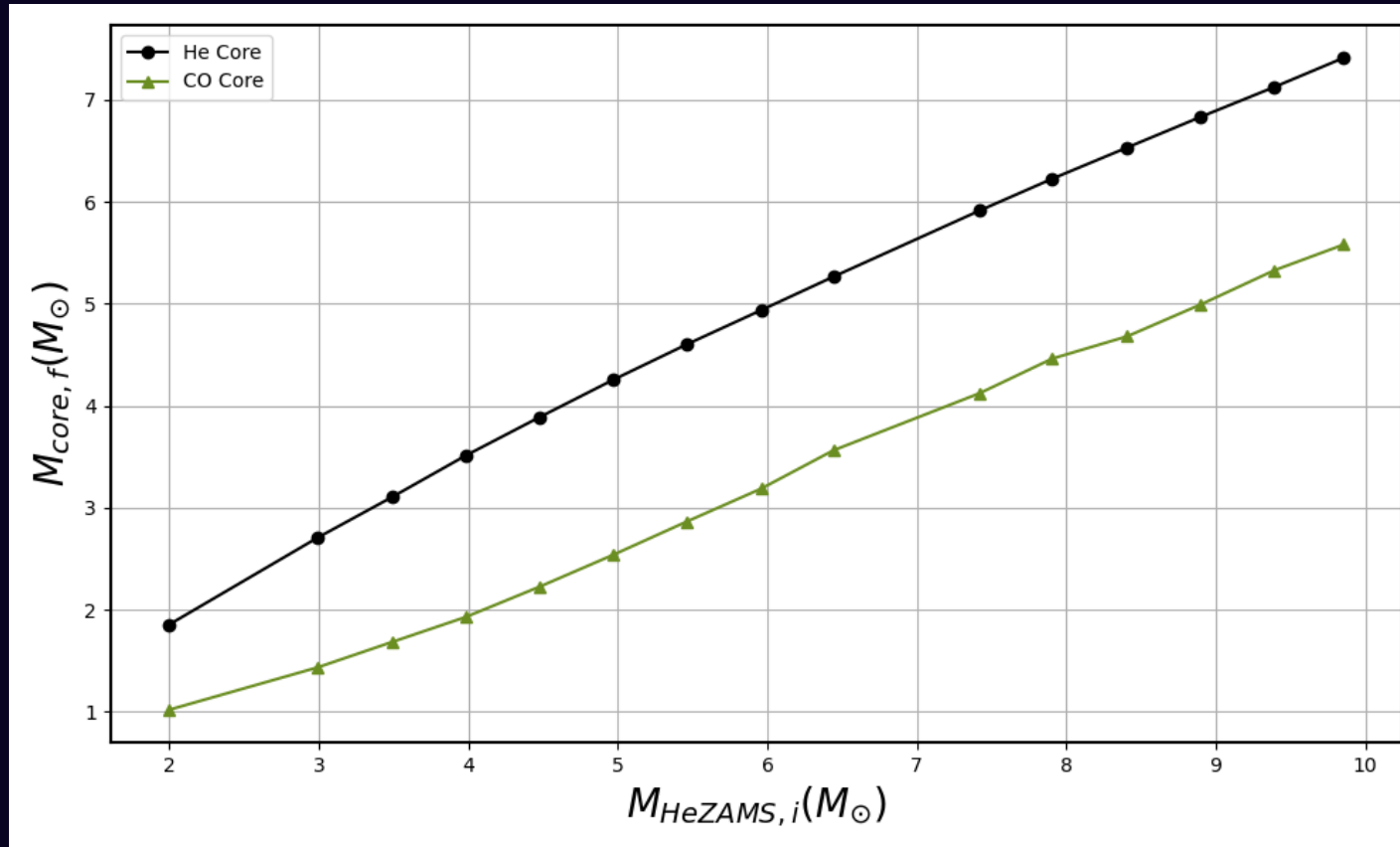
- Aiming to calibrate the model by ensuring it can explain the galactic DNS population
- Use that model to examine predictions for GW - compare different proposed pathways for GW190425 like systems
- This work roughly explain the galactic DNS population, but only a small fraction of DNS form massive GW190425 like systems – they come from intermediate mass helium star $5M_{\text{sun}} - 7.0M_{\text{sun}}$
- Since this work can't explain a high rate of massive gw190425 like event (given we have only detected 2 events and 1 massive event)in this model.
- Future work will include exploring grid of models with low metallicity, massive primary NS and super-Eddington accretion.

Extra slides

HRD of $3.4 M_{\odot}$ He + $1.4 M_{\odot}$ NS binary

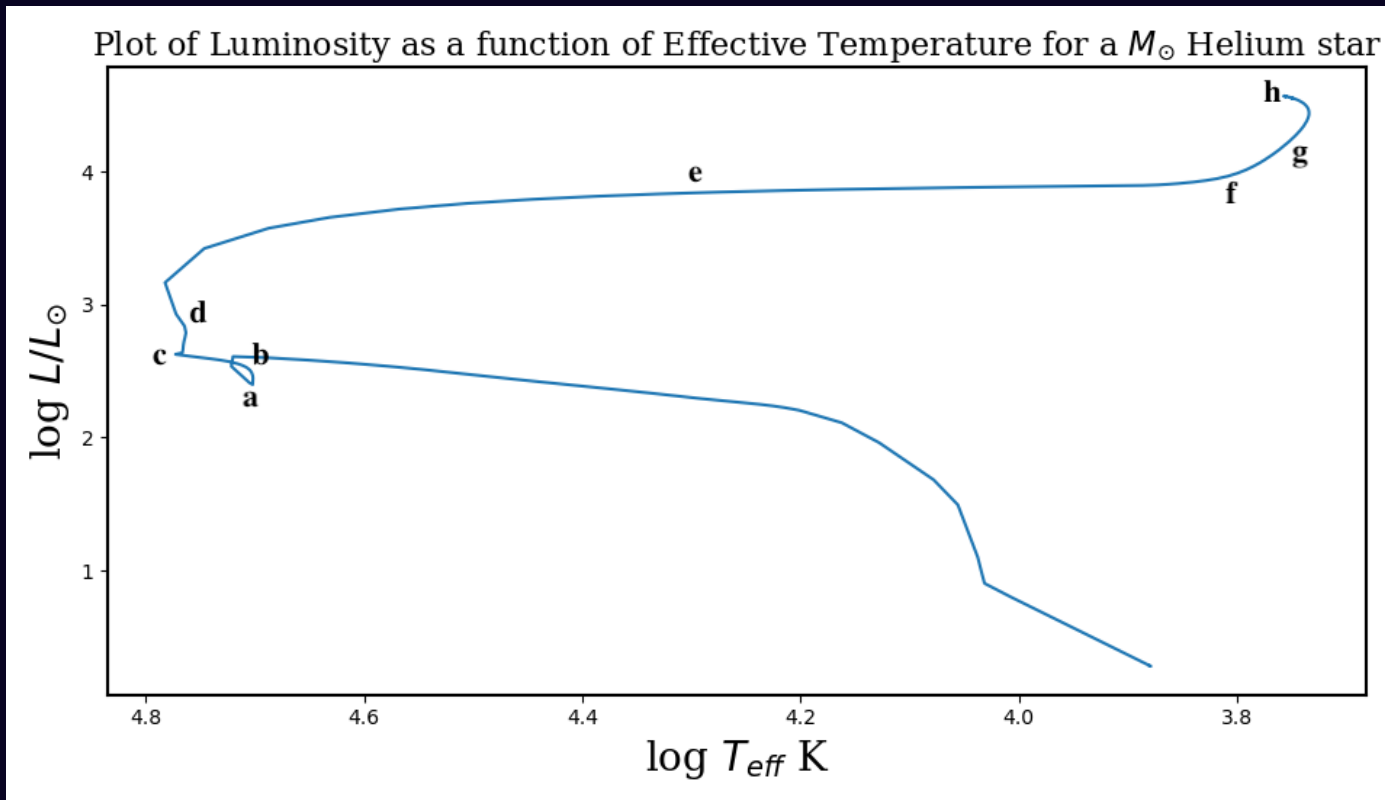


He and carbon-oxygen (CO) core masses of all He-stars



- Core masses are well determined by the star's initial mass and standard choice of stellar physics.

1 $M_{\odot}$ Helium Star Evolution



a : zero age main sequence

a - b : Main sequence phase

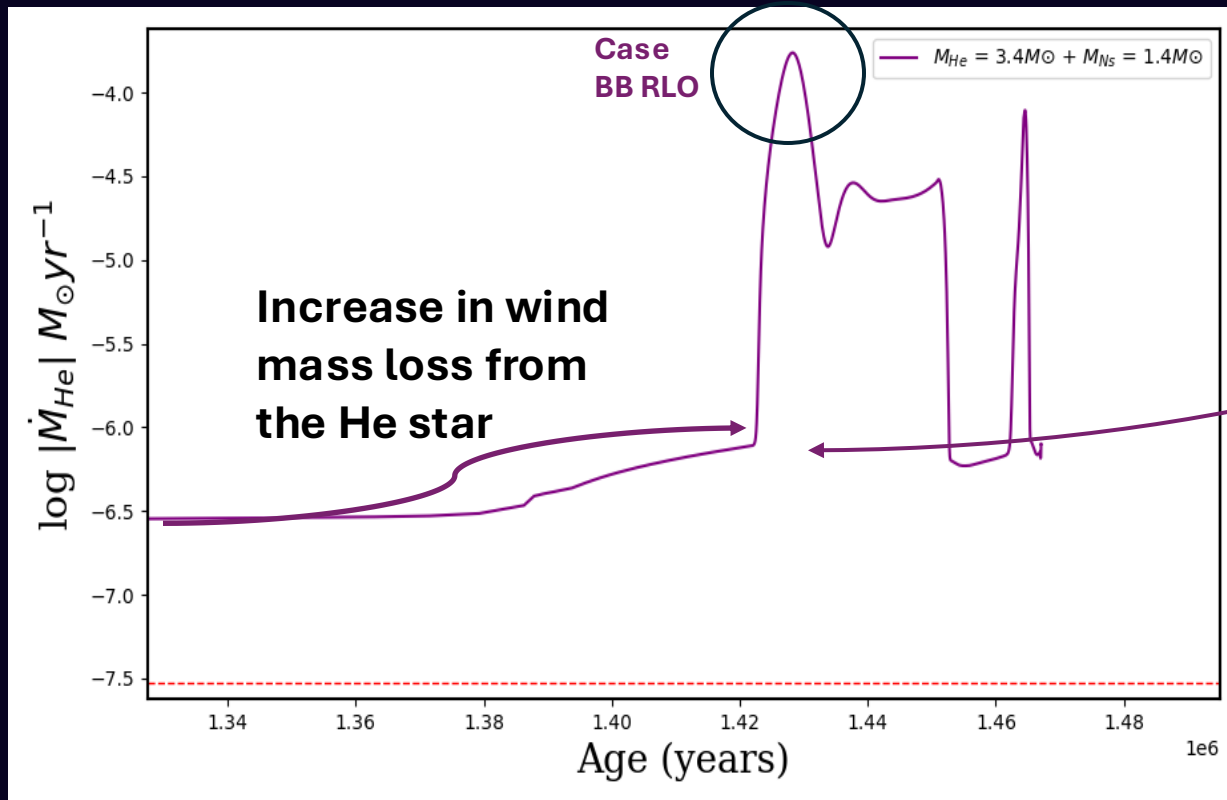
b - c : exhaustion of central helium,

c - d : first cooling phase,

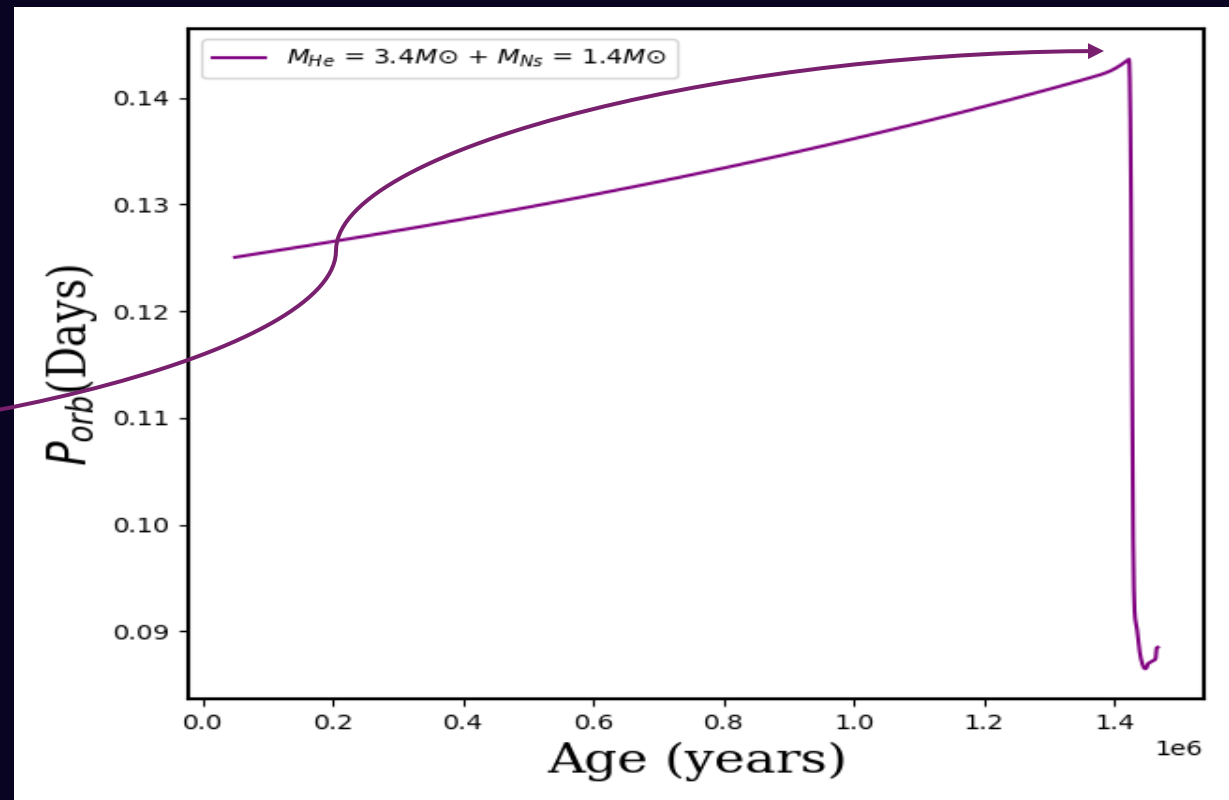
d - e : slow phase of shell helium burning

e - h : fast phase of helium shell burning

Wind mass loss from the He star as a function of the star's age



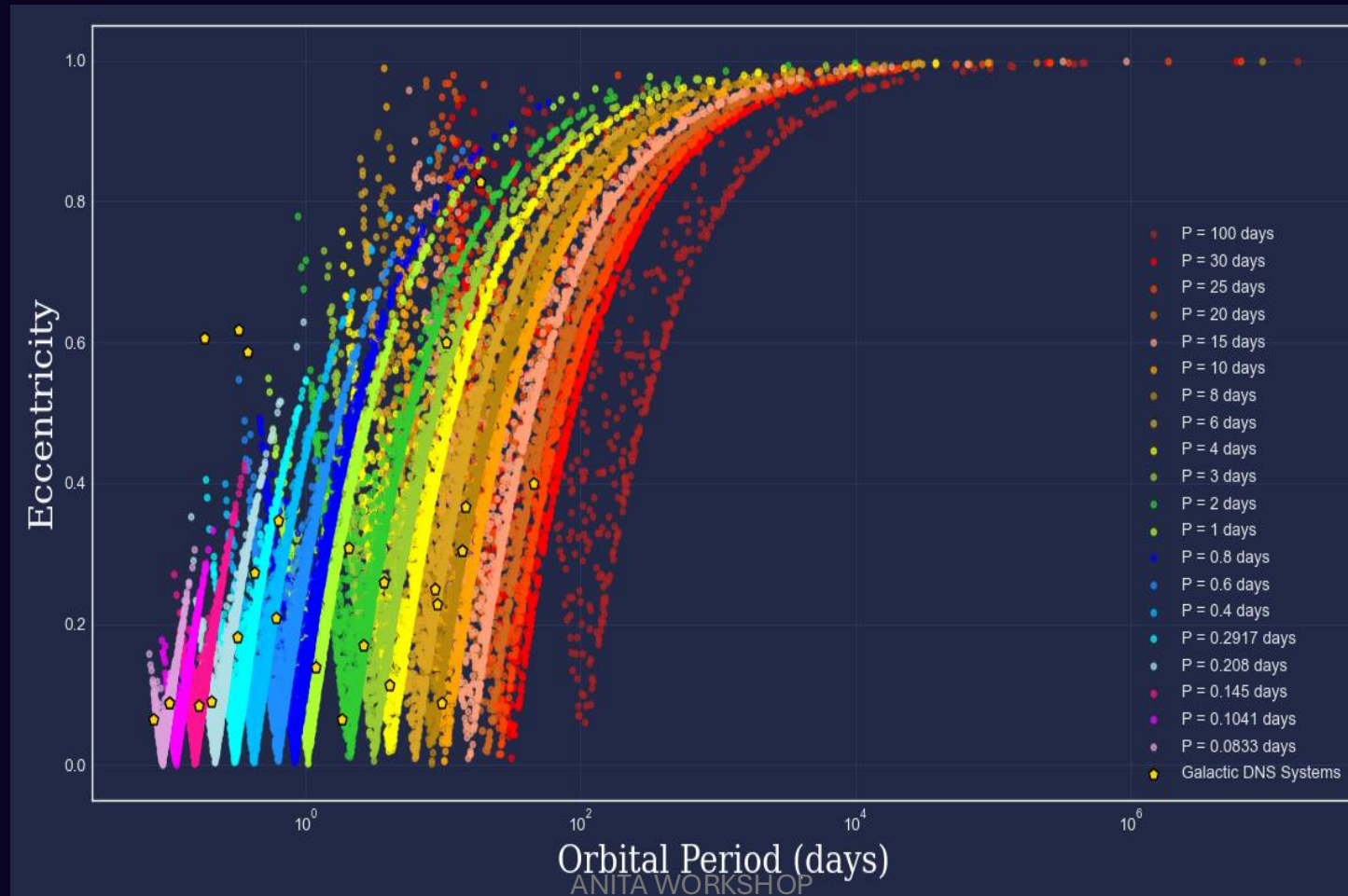
Change in orbital period as a function of the star's age



Eccentricity as a function of orbital period for all confirmed DNS systems observed in the Galactic disk (yellow pentagons) and simulated systems.

Without Ultra-stripped Supernova (USSN) and Electron Capture Supernova (ECSN) Condition

$$M_{\text{He},i} = 3 M_{\odot}, M_{\text{CO}} = 1.33\text{-}1.44 M_{\odot}$$



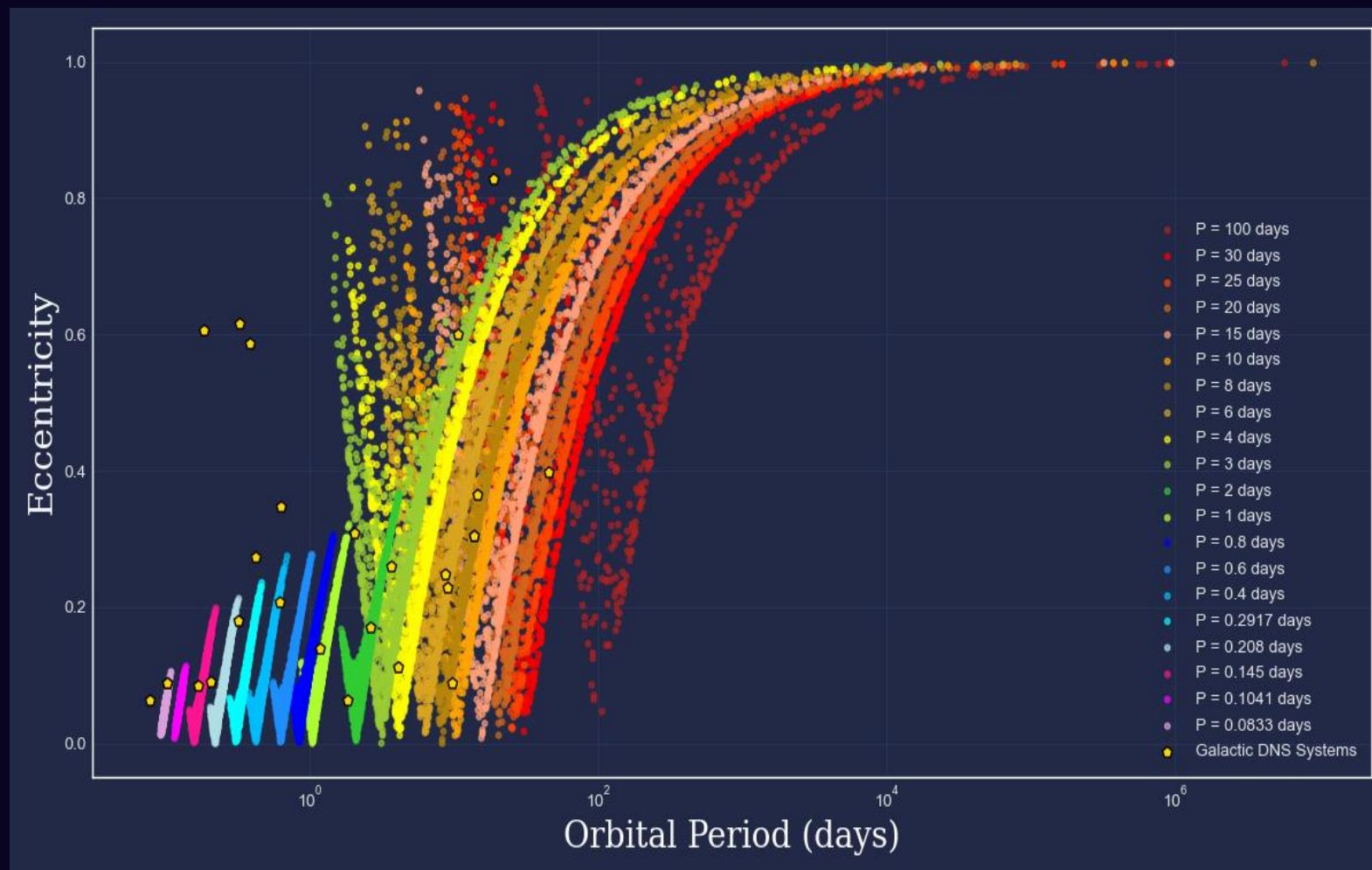
After incorporating USSN and ECSN together

USSN

$$M_{\text{He env}} \leq 0.2 M_{\odot}$$

ECSN

$$1.37 M_{\odot} < M_{\text{CO}} < 1.43 M_{\odot}$$



μ_{2a}	μ_{2b}	V_{ns} (km/s)	Fraction of Heavy DNS (%)	
			Radio	GW
1.4	0.5	400	16.00	18.80
1.4	0.5	525	14.99	17.22
1.4	0.5	550	14.75	17.03
1.4	0.3	400	8.08	9.55
1.4	0.3	525	7.24	8.83
1.4	0.3	550	7.04	8.71
1.4	0.1	400	1.96	2.84
1.4	0.1	525	2.02	2.93
1.4	0.1	550	2.03	2.96
1.35	0.5	400	12.36	14.60
1.35	0.5	525	11.61	13.58
1.35	0.3	400	4.84	6.13
1.35	0.3	525	4.37	5.78
1.3	0.5	400	9.31	11.19
1.3	0.5	525	8.42	-
1.3	0.3	400	2.84	3.86
1.3	0.3	525	2.77	2.82

Table 1: Fraction of heavy DNS for different mass parameters and kick velocities.